

Armaan Sandhu

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Education

Worcester Polytechnic Institute | *Worcester, MA* 2025 - Present
B.S. in Computer Science, M.S. in Artificial Intelligence | **GPA: 4.0** | Expected May 2029
Carmel Junior College | *Jamshedpur, India* | **Valedictorian** 2010 - 2025

Technical Skills

Languages: Python, C, C++, Rust, Zig, Java, x86-64 Assembly, JavaScript
Tools: Git, GDB, Linux, Bash, Valgrind, llama.cpp, Redis, PostgreSQL, SQLite
Concepts: OS Kernel Development, Systems & Concurrent Programming, Binary Analysis
AI/LLM: MCP, LLM Routing & Gateways, RAG, Vector Search, Tokenizers, Spec Driven Development

Open Source Contributions

AI Infrastructure (LiteLLM, LanceDB, Lance) May 2026 - July 2026

- Fixed production data-loss and request-cancellation bugs and added universal gateway support for several AI agent harnesses

Systems & Browser Engines (systemd, CPython, Lightpanda) May 2026 - Present

- Nearly **20 merged systemd PRs** fixing memory-safety, networking, parsing, and security issues; developed a journalctl optimization delivering **1.7-2.5×** faster reads.
~40 pull requests merged across 6 major open-source repositories, spanning Rust, Python, Zig, and C.

Experience

Founding Engineer & Product Head | *MLRouter* June 2026 - July 2026

- Built an OpenAI-compatible gateway for 400+ models with cost/latency routing, provider failover, and enterprise governance; took it from concept to production.

Open Source Author & Maintainer | *mcp-persist* May 2026 - Present

- Built and maintain a production-grade MCP persistence layer across Redis, SQLite, and PostgreSQL, reaching **10k PyPI downloads in its first month** and 600+ tests.

Research Experience

Independent Study, RunixOS | *WPI, advised by Prof. Craig Wills* Early 2026 - July 2026

- Built a capability-based OS kernel from scratch, implemented a preemptive scheduler, synchronous / asynchronous IPC, and capability-gated filesystem to study OS design under capability security.

Projects

codebase-rag (Python) July 2026 - Present

- Built a codebase intelligence tool for AI coding agents that answers cross-file questions in a single query and automatically keeps its index synchronized with code changes.

rag-timetravel (Python) June 2026 - Present

- Built the core debugging engine behind codebase-rag, letting developers rewind searches to the exact data version used, replay queries, and pinpoint why results changed.

TokenDrift (Python) June 2026 - Present

- Built a model-upgrade safety check that measures shifts in token counts, costs, and vocabulary before changes reach production, now integrated into MLRouter.

Binary Exploitation (C, x86-64 Assembly) | *WPI, CS2011 Machine Organization & Assembly*

- Reverse-engineered a multi-stage binary and its testing harness from scratch, using GDB and objdump to understand and reproduce each exploit.

Cache Simulator (C) | *WPI, CS2011 Machine Organization & Assembly*

- Built an N-way set-associative cache simulator with LRU eviction and used its output to drive and validate matrix-transpose optimizations.